

BRAF fusion-hotspot benchmark report

Study: msk_impact_2017

Abstract

This report analyzes BRAF gene fusions from the msk_impact_2017 study, covering 41 fusion events. 33/41 fusions (80.5%) were in-frame. 33/41 fusions (80.5%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.00060718$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 80.5% in-frame and 80.5% domain retention.

1 Results summary

params['group_field'] is required

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 41 analyzed fusion events, identifying 25 distinct fusion partner genes. The most frequent partner was SND1, observed in 8 events.

1.6 Composite evidence score

Composite evidence score reported: confidence_certainty
sub-score excluded: no applicable confidence_stats
MLE/CI result was supplied.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 31/33 (93.9%) of in-frame fusions retained the domain, $p=0.00060718$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

1.7 Exon retention

Exon retention reported: No target exon was supplied in
params or GeneConfig.

1.3 Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 31/33 (93.9%) of in-frame fusions disrupted the domain, $p=0.0049969$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.0249975.

1.8 Joint partner

Joint partner reported: Algorithm failed: ValueError:
joint_partner requires a GeneConfig with gene_pair

1.4 Cutpoint detection

Cutpoint detection scanned 40 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 439 aa (permutation-corrected $p=0.0599401$, not statistically significant at $\alpha=0.05$). This is 19 aa from the nearest configured domain boundary at 458 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: Algorithm failed: ValueError:

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
SND1	8	0.1951219512195122	0.30004340774793187	0.1602103418604825	0.759046052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3104960150774111	1
SCRIB	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29374443719979015	2
PJA2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.993421052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29264794597172006	3
RBM33	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.993421052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29264794597172006	4
KIAA1549	4	0.0975609756097561	0.30004340774793187	0.1602103418604825	0.7105263157894737		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2698890670671415	5
CDK5RAP2	2	0.04878048780487805	0.30004340774793187	0.1602103418604825	0.805921052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.269528027272533	6
GIPC2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24330584070856212	7
KDM7A	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24330584070856212	8
OSBPL9	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24330584070856212	9
PRKAR2B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24330584070856212	10
ZNF207	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.6973684210526316		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24330584070856212	11
AGAP3	2	0.04878048780487805	0.30004340774793187	0.1602103418604825	0.6151315789473684		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2377297816584979	12
CCDC6	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.618421052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23014794597172	13
FAM131B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.618421052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23014794597172	14
PHTF2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.618421052631579		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23014794597172	15
CUL1	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22905145474364985	16
LINC00244	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22905145474364985	17
METTL2B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22905145474364985	18

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
PRKAR1B	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.611842105263158		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22905145474364985	19
AGK	3	0.07317073170731707	0.30004340774793187	0.1602103418604825	0.26315789473684215		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18719758225755653	20
MKRN1	3	0.07317073170731707	0.30004340774793187	0.1602103418604825	0.19956140350877194		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1765981670528782	21
MRPS33	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.26315789473684215		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17093741965593054	22
VIPR2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.17105263157894735		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15558654246294806	23
PARP12	1	0.024390243902439025	0.30004340774793187	0.1602103418604825			['domain_disruption', 'domain_retention', 'recurrence']	0.1524933246397482	24
LUC7L2	1	0.024390243902439025	0.30004340774793187	0.1602103418604825	0.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12707777053312352	25

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
287	0	33	1	6	0.0	0.17500000000000004	0.7569619513137055
327	3	30	2	5	0.25	0.20399751978699343	0.6903751127180751
380	11	22	3	4	0.6666666666666666	0.6789250550860457	0.168178163803398
381	20	13	4	3	1.1538461538461537	1.0	-0.0
393	25	8	4	3	2.34375	0.36939377457953293	0.43251042803096634
438	30	3	5	2	4.0	0.20399751978699346	0.690375112718075
439	33	0	5	2		0.026923076923076925	1.569875307956561
565	33	0	6	1		0.175	0.7569619513137056

2.3 domain_disruption frame_domain_contingency_table

31	3
2	4

2.4 domain_disruption domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
ras_binding	RAS-binding domain	40	6	0	34	0.0

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
cysteine_rich	Cysteine-rich domain	40	6	0	34	0.0

2.5 domain_retention frame_domain_contingency_table

31	2
2	5

2.6 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	40	33	1	6	0.06050420168067227

2.7 frequency Partner_gene_counts

Partner_gene	Event_count
AGAP3	2
AGK	3
CCDC6	1
CDK5RAP2	2
CUL1	1
FAM131B	1
GIPC2	1
KDM7A	1
KIAA1549	4
LINC00244	1
LUC7L2	1
METTL2B	1
MKRN1	3
MRPS33	1
OSBPL9	1
PARP12	1
PHTF2	1
PJA2	1
PRKAR1B	1
PRKAR2B	1
RBM33	1
SCRIB	1
SND1	8
VIPR2	1
ZNF207	1

2.8 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_protein	breakpoint_4	domain_retained	domain_deleted	domain_retained	domain_deleted	disrupted_domains	source_annotation	stop_effect_on_fusion
EVT-P-0009757-T01-IM5-1	P-0009757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAFInv Protein fusion: in frame (AGAP3-BRAF)	in frame
EVT-P-0007861-T01-IM5-2	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAFInv Protein fusion: in frame (AGAP3-BRAF)	in frame
EVT-P-0000913-T02-IM5-3	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFInv Protein fusion: in frame (AGK-BRAF)	in frame
EVT-P-0010945-T01-IM5-4	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12648:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	in frame
EVT-P-0004477-T01-IM5-5	P-0004477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10078:AGK_c.981-1138:BRAFInv Protein fusion: in frame (AGK-BRAF)	in frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_fusion	is_deleted	description	annotation	effect_on_fra
EVT-P-0004032-T01-IM5-27	P-0004032-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487892	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 in frame (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)		
EVT-P-0009097-T01-IM5-28	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 in frame (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)		
EVT-P-0003664-T01-IM5-29	P-0003664-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482817	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		c.4929+549:KIAA1549_c.1314+4:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)		
EVT-P-0006412-T01-IM5-30	P-0006412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel Antisense fusion NA NA		
EVT-P-0002305-T01-IM3-31	P-0002305-T01-IM3		METTL2B-BRAF	METTL2B	unknown	three_prime	140491674	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA NA		
EVT-P-0004373-T02-IM5-32	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)		
EVT-P-0001036-T01-IM3-33	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA in frame		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	domain_retained	domain_disrupted	description_annotation	effect_on_fra	
EVT-P-0007533-T01-IM5-46	P-0007533-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482226	10	438	True	retained	1.0	False	Protein kinase domain RAS-binding domain; Cysteine-rich domain	SND1 (NM_014390) - BRAF fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738: Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-0001687-T01-IM3-47	P-0001687-T01-IM3		SND1-BRAF	SND1	in-frame	three_prime	140490224	9	381	True	retained	1.0	False	Protein kinase domain RAS-binding domain; Cysteine-rich domain	SND1 (NM_014390) - BRAF fusion: inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-0008920-T01-IM5-48	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	9	393	True	retained	1.0	False	Protein kinase domain RAS-binding domain; Cysteine-rich domain	ZNF207 (NM_001098507) - BRAF rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.7548117::) Protein fusion: in frame (ZNF207-BRAF)	in frame

3 Figures

BRAF domain-retention track

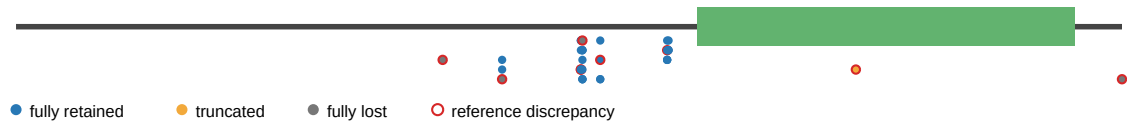


Figure 1: domain retention outliers

Reference vs msk_impact_2017



Figure 2: reference comparison